ST456 Group Project - deep-strom

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Abstract

We aim to train and fine-tune deep learning architectures on the BirdCLEF ecological dataset for automatic bird sound recognition and classification, motivated by the need for accurate and scalable biodiversity monitoring. We begin with convolutional neural networks (CNNs) to extract features from mel spectrograms and iteratively extend our models with recurrent (BiGRU and BiLSTM) and attention-based mechanisms to capture temporal dependencies. SpecAugment – with time cropping, dynamic time/frequency masking, Gaussian noise, and padding (instead of resizing) – proves critical in reducing overfitting and improving generalization. Unfreezing ResNet50's conv3-5 blocks and incorporating a hybrid RNN+attention head leads to a 72% performance improvement over the baseline ResNet50 (34% $\rightarrow$ 60%). We find that integrating rich metadata (e.g., latitude/longitude) further improves accuracy. Extreme class imbalance in the dataset means common metrics like CMAP@3 underweight rare classes. Focal loss weighted by class frequency (α = class weights) records a CMap loss and doesn't outperform on minority classes. Oversampling outperforms focal loss on the same architecture and improves recall on 10 classes .Cosine similarities reveal a high degree of temporal overlap between spectograms of confused bird callsso we explore transformer approaches but data quality and class imbalances limit performance upside. Finally, by transitioning from primary label classification to multi-label training (reflecting multiple species per clip), we achieve a peak CMAP of 81.36%.

1. Introduction

Biodiversity monitoring is essential to understanding and protecting ecosystems. Birds are widely recognized as key indicators of wildlife health, and passive acoustic monitoring provides a scalable method for tracking bird populations, although challenges remain due to variability in natural sounds and manual validation requirements (Navine et al., 2024). Recent advances in deep learning offer promising solutions to automate bird sound recognition (Stowell et al., 2019; Kahl et al., 2021). Neural networks can learn rich representations from audio data, improving both the speed and accuracy of classification tasks (Hershey et al., 2017). We aim to build an efficient system that transforms raw audio recordings into machine-readable features and trains neural networks to predict bird species from environmental soundscapes. We focus on three core challenges: 1. Handling real-world audio with significant noise. 2. Extracting robust features for input to deep learning models. 3. Ensuring generalization across diverse species and recording conditions.

2. Related Work

Automatic acoustic monitoring has become an increasingly important area of research in ecological and biodiversity studies. Early work by Stowell et al. (2019) highlighted the challenges associated with bird audio detection and demonstrated that deep learning models, including CNNs and RNNs, could significantly outperform traditional methods based on handcrafted audio features. Their "Bird Audio Detection Challenge" established a benchmark for applying deep learning to noisy and complex field recordings. Building on this foundation, Kahl et al. (2021) developed BirdNET, a large-scale bird species classification system leveraging deep neural networks. BirdNET demonstrated that deep learning architectures could achieve high accuracy amidst diverse environmental conditions. In parallel, Hershey et al. (2017) explored the application of CNN architectures, originally developed for image classification, to large-scale audio classification problems. By applying CNNs to spectrogram-like representations of audio signals, they demonstrated that deep convolutional models could effectively learn rich time-frequency features directly from raw audio data. Together, these studies motivate our approach to employ deep learning models trained on spectrogram representations to classify bird species from environmental audio recordings.

We build on established foundations – CNNs on log-mel spectrograms (Kahl et al., 2021; Hershey et al., 2017) and BiRNNs for temporal modeling (Stowell et al., 2019) – but push further. Our hybrid architecture fuses ResNet50 with a hybrid BiGRU-BiLSTM and self-attention, capturing both

local and global acoustic structure. Where prior work optimizes for accuracy at scale, we reorient toward conservation recognising importance of rare species. Focal loss ($\gamma = 2.0$, $\alpha =$ class weights) actively attempts to resurface rare species. Oversampling succeeds in imporving performance on minority classes. Confusion analysis between dominant and underrepresented classes (e.g., Cinnamon Bracken-Warbler vs. African Gray Flycatcher) adds translated actionable insight. We extend standard augmentations (e.g., SpecAugment) with dynamic time cropping. Finally, our multi-label adaptation (81% CMAP@3) tackles overlapping birdcalls in single audio file, a field condition underrepresented in most benchmarks. In a field shaped by statistical dominance, our work attempts to account for ecological priority.

3. A Note on 3 Data Limitations and Modelling choices

A core challenge – and defining feature – of this dataset is its extreme class polarization: some species appear in fewer than two samples, others in over 350 (Figure 5 - boxplot). However our use-case focused on conservation values rare species calls. While we value the ecological importance of rare species, transformer models demand context. With < 2 examples, positional embeddings cannot stabilize and self-attention has no variation to learn from. Industry standard : ImageNet enforces 398 samples/class, AudioSet 85. Our cutoff (≥ 2) is lenient, supported by augmentation and pretrained transfer. As Kahl et al. (2021) found, classes with < 5 examples degrade accuracy by up to 17%. Further the original .ogg audio files vary in length typically 30s to 1 min and we split into 5s clips for spectrogram generation. Statistically, most 5s windows from a file labeled *afrsil1* (African Silverbill) will still feature that species. CNNs and transformers, supported by global pooling, learn to focus on dominant acoustic patterns and filter transient noise. However we also risk cropping bird calls – to mitigate this we introduce randomised time cropping to simulate shortened calls. Further, the inherent raw data maps each audiofile to a single 'primary label' representing the dominant species, though each file contains several overlapping calls. We view this as a key shortcoming of the data and later mine metadata for 'secondary classifications' and shift toward multi-label classification allowing the model to account for co-occurring species in a single audio clip.

4. The Architecture

We report on three key architectures that achieved significant performance gains, building on lessons from prior iterations and data mining that shaped each design. For completeness, we append a summary of eight total models at the end of the report (Table 1) and include all corresponding models in code files.

4.1. A First-Principles Aside on CNNs

Multilayer perceptrons (MLPs) perform well on tabular data. But when applied to image data, like spectrograms, flattening the input would destroy spatial relationships and explode the number of parameters. Convolutional neural networks (CNNs) are designed with spatial invariance in mind, exploiting the fact that nearby pixels are often correlated.

A standard fully connected layer computes a weighted sum over all input units for each output. CNNs instead apply convolutional filters – small kernels that slide across the input, focusing on local spatial neighborhoods. This is made possible by weight sharing and relative indexing: rather than learning a separate weight for each position, the same weights are used across different positions in the input. Mathematically, this translates to:

$V_{i,j,k,l} = w_{i,j,i+k,j+l}$, and if we assume weights are independent of i, j, we get $V_{k,l}$, the standard convolution kernel. This reduces parameters, improves generalization, and enables translational equivariance, the ability to recognize patterns regardless of location.

Key components:

- **Padding** preserves border information during convolutions.
- **Strides** downsample spatial dimensions while reducing computation.
- **Channels** allow for multi-feature representations (e.g., RGB for images). Our mel spectrograms are grayscale (1-channel), but are repeated to 3 channels to fit pretrained model input formats.
- **Pooling layers** (e.g., max pooling) introduce translational invariance and reduce dimensionality. They ensure that features remain robust even with slight positional shifts.

4.2. Why ResNET50?

We chose ResNet50 as a baseline due to its consistent performance on visual recognition benchmarks like ImageNet, particularly for Top-1 and Top-5 accuracy (see Figures 4). ResNet introduces the concept of residual learning (He et al., 2016). Instead of learning a full transformation H(x), each residual block learns $F(x) = H(x) - x$, so the model effectively learns $H(x) = F(x) + x$. If no change is needed, $F(x)$ becomes zero, and x passes through unchanged. This formulation allows very deep networks to train without suffering from vanishing gradients. Each residual block in ResNet50 contains two 3×3 convolutional layers, batch normalization, and ReLU activations, with a skip connection that adds the input directly to the output. Unlike VGG,

ResNet does not use dropout; batch normalization is sufficient for regularization. During training, images (in our case, spectrograms) are resized to 224×224 and optionally augmented. By training only the residuals, each block makes fine-grained adjustments rather than relearning redundant representations, allowing deeper architectures to converge efficiently.

4.3. Why ResNET50 Avoids the Vanishing Gradient Problem

In deep neural networks, backpropagation relies on the chain rule to compute gradients. For an input x and a final loss L, we calculate:

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial a_n} \cdot \frac{\partial a_n}{\partial a_{n-1}} \cdot \frac{\partial a_{n-1}}{\partial a_{n-2}} \cdots \frac{\partial a_1}{\partial x}$$

Where ai is the activation at layer i. The problem arises when each derivative term involves the gradient of an activation function like sigmoid or tanh, whose gradients are bounded between 0 and 1. As these are multiplied across many layers, the overall gradient can shrink exponentially:

$$\left| \frac{\partial L}{\partial x} \right| \to 0$$

This is the vanishing gradient problem: early layers learn extremely slowly or not at all.

4.4. How ResNET Helps

ResNet mitigates this issue through residual connections. In a residual block, instead of learning the transformation $H(x)$, the network learns the residual $F(x)$, so:

$$H(x) = F(x) + x$$

This leads to an output:

$$y = x + F(x)$$

During backpropagation, the gradient becomes:

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} \cdot \left(\frac{\partial F(x)}{\partial x} + I \right)$$

Where I is the identity matrix from $\frac{\partial L}{\partial x}$. Even if $\frac{\partial F(x)}{\partial x}$ is small, the identity term ensures that some gradient still flows through the network. This allows deeper architectures to learn effectively without vanishing gradients.

At the final layer, a softmax activation is applied to generalize the logistic function across dimensions. This converts the model's raw outputs into a probability distribution across all bird species classes:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

This output is used to compute cross-entropy or focal loss, depending on the architecture variant.

4.5. Model 1: ResNET50 + BiGRU-BiLSTM + Multi-Head Self Attention

We begin with a ResNet50 backbone initialized with frozen ImageNet weights and a simple classification head consisting of global average pooling, dropout, and a dense softmax layer. This baseline achieved a validation CMAP@3 of 34.63%. However, significant overfitting was observed, with training performance diverging sharply from validation metrics.

To mitigate overfitting, we introduce SpecAugment and carefully tailored preprocessing. Key augmentations include:

- Time cropping to preserve the structure of calls truncated during 5-second segmentation.
- Padding instead of resizing to avoid spectral distortion.
- Dynamic time and frequency masking, with masking widths set as proportions of the input dimensions (up to 10%).
- Additive Gaussian noise, sampled from $\mathcal{N}(0, \sigma^2)$, to simulate recording variation.

While these changes didn't see improved CMap@3 they rectified overfitting with validation set outperforming training data. Recognizing that mel-spectrograms encode sequential acoustic patterns, we introduce temporal modeling via RNNs. Unlike text, bird calls exhibit bidirectional dependencies; future notes often disambiguate prior sounds. Accordingly, we stack:

- A Bidirectional GRU, suited for short-term transients,
- Followed by a Bidirectional LSTM, capable of retaining long-range harmonic structure.

To enhance focus on diagnostic audio segments, we add a multi-head self-attention layer with 8 parallel heads. A residual connection and layer normalization follow to preserve the signal and stabilize training. Dropout (0.4) further combats overfitting across > 800 RNN units. Finally, we progressively unfreeze the `conv5_x`, `conv4_x`, and `conv3_x` blocks of ResNet50. This allows the CNN backbone to adapt not only high-level representations but also lower-level frequency edge detectors to the bird domain.

Result: This hybrid architecture – CNN + RNN + attention – boosts CMAP@3 to 60%, representing an 72% improvement over the baseline frozen ResNet50.

Continuation Matrix

At this stage, we analyze the model's misclassifications using a confusion matrix (Figure 2,3). Though a neat-diagonal emerges reflecting strong model3 performance, further analysis reveals a consistent trend: rare species ($< Q1$) are

often misclassified as more prevalent species (≥ 100 samples) (Table 2). This systematic bias suggests the model is underperforming on classes relevant to our conservation-driven use case, where accurate identification of rare species is critical.

For instance:

- African Gray Flycatcher (6 samples) is frequently confused with Cinnamon Bracken-Warbler (79 samples),
- Hinde's Pied-Babbler (6 samples) is misclassified as Woodland Kingfisher (47 samples).

To address this, we replace the standard categorical cross-entropy loss with focal loss, which better handles class imbalance. We use:

- High gamma ($\gamma = 2.0$) to down-weight easy examples and focus learning on harder, underrepresented classes,
- Alpha weighting proportional to class inverse frequency, ensuring rare classes contribute more to the gradient.

This yields limited results, with only 1 class improving and 96% stagnating on recall. However, notably introducing oversampling surpasses focal loss methods and outperforms hybrid CNN-RNN Model3 improving CMap to 63.18 (3.18 pct pt improvement from model3) and $\sim$10 pct pt improvement on focal loss). It also improves recall on 10 species (Figure 7).

For completeness, to investigate whether the confusions stem in part from intrinsic acoustic similarities between confused species, we compute spectrogram-level cosine similarity between the most confused class pairs with $>$90% similairity in 100% confused species (Table 4). We also eyeball similarities between confused-species' spectrograms (Figure 6), Eg : both the african gray flycatcher and cinnamon bracken warbler feature short bursts of repeated trills followed by pauses. This analysis supports the hypothesis that some confusion arises from overlapping call patterns. An insight that motivates the Transformer-based architectures explored in subsequent models. Lastly, we explore EfficientNetB0 (5.3M parameters) and EfficientNetB3 (12M parameters) as lighter alternatives to ResNet50 (Tan & Le, 2020). However, despite their computational efficiency, the EfficientNetB0-based model does not outperform Model 1 in CMAP@3, while EfficientNetB3 outperforms only slightly.

4.6. Model 2: Transformer-Based Architecture

In this model, we implement a Transformer-based architecture inspired by Vision Transformers (ViTs) to process Mel spectrograms. Instead of relying on convolutional inductive biases, this architecture tokenizes the spectrograms and uses self-attention to model long-range dependencies in time and frequency.

The core components of the architecture include:

- Linear Projection ($input_proj$): Each 128-dimensional frequency frame is projected into a 512-dimensional embedding space.
- CLS Token: A learnable classification token is prepended to the sequence and later used for final prediction via cross-attention.
- Positional Encoding: Sinusoidal positional embeddings are added to preserve temporal structure across the spectrogram.

This design allows the model to capture global time-frequency interactions more flexibly than CNNs, which rely on spatial locality. However, we note that bird audio has strong local temporal regularities, so the benefits of global modeling must be weighed against architectural complexity.

For optimization, we use:

- AdamW optimizer with lr=1e-3 and $weight_decay$=1e-4, as it decouples weight decay from adaptive learning rate, is better suited for attention-heavy models.
- CosineAnnealingLR scheduling (T_max=30, eta_min=1e-6), which smoothly decays the learning rate over time, allowing systematic exploration of the parameter space.
- Early stopping (patience = 5) to avoid overfitting and reduce unnecessary computation.

This configuration draws on standard best practices in Transformer-based modeling. While the theoretical flexibility is promising, we find that practical gains were limited, likely due to the relatively small dataset size and the highly structured nature of spectrogram inputs. Nonetheless, this model provides a valuable benchmarking framework and informs further hybrid designs.

4.7. Model 3: Multi-Label Spectrogram Classificaiton with ResNeSt-50d Backbone

Drawing from everything we've learnt so far, we experiment with mining metadata for secondary class assignments. In section 3 on data we identified the potential drawbacks of assigning one umbrella class to an audio file with multiple bird calls and the associated risk of data loss.Though sparse and with a high percentage of nulls the metadata contains

a 'secondary' column, we rerun our training using added features.

We note that here also we increase the data available per class but this is very different to oversampling, instead of repeated instances, we now have all new data on minority classes.

The implemented model builds on ResNeSt-50d, a state-of-the-art convolutional backbone that extends ResNet via split-attention blocks, enabling better channel-wise feature recalibration. We adapt this architecture for single-channel melodic spectrograms, making it particularly well-suited for the multi-label bird sound classification challenge presented by BirdCLEF. Key design elements include:

- Temporal Attention Pooling: Enables the model to focus selectively on time regions with informative acoustic features, suppressing background noise or silent intervals.
- SpecAugment + Mixup: Time-frequency masking and interpolative augmentation improve generalization across call durations and acoustic environments.
- Sliding Window Cropping: Ensures that varying call lengths are captured without truncation or padding artifacts.

The dataset is stratified into train, validation, and test splits, with labels reformatted into multi-label binary vectors. We use a weighted random sampler to mitigate extreme class imbalance. Our loss function is sigmoid focal loss ($\gamma = 2.0$), which down-weights easy negatives and sharpens the model's focus on rare and underrepresented species. Optimization was done using Adam with a OneCycle Learning Rate Scheduler and Cosine Annealing from 1e-3 $\rightarrow$ 1e-6 over 30 epochs. Precision scaling via PyTorch AMP (Automatic Mixed Precision) accelerated training while preserving numerical stability. Early stopping (patience = 5) was used to halt training when validation cmAP plateaued. Performance Summary:

- Best epoch: 28
- Validation cmAP: 0.8084 (up from 0.5864)
- Validation Hamming Accuracy: 0.9975
- Final loss: 0.0001 (train and val)

These results highlight stable convergence, strong generalisation, and high multi-species detection accuracy. In particular, the dramatic improvement in cmAP – especially for rare classes – validates our architectural choices and training regime. Taken together, this model represents a viable, high-performance pipeline for real-world, conservation-grade bird sound classification.

5. Training methods

A description of the training process, including optimisers, hyperparameter settings, and techniques to mitigate overfitting.

5.1. OPTIMISERS

A brief overview of common optimisers to justify the choice of Adam is outlined :

1. SGD updates weights according to the gradient of the loss function, however, it can get stuck in local minima especially when functions have non-convex loss surfaces. Convergence is an issue with multivariate unconstrained optimization. The following are adaptive learning rates in SGD – ie they dynamically adjust the learning rate, allowing the algorithm to modify the update size for each parameter depending on its past gradients.
2. 'AdaGrad': uses the inverted sq root of diagonal elements of a matrix that captures the sum of historic gradient at each training epoch. (the off-diagonal elements capture interactions between parameters and it would be computationally expensive to use the whole matrix).
3. 'RMSProp': AdaGrad weights all elements equally regardless of how far back they occured, RMSProp uses exponential moving average and assigns greater weights to more recent gradients. Prevents LR from decreasing too quickly.
4. Nesterov Accelerated Gradient (NAG) is an improved type of momentum-based SGD. Instead of updating weights immediately, it first moves slightly in the previous direction (momentum),then calculates the gradient at that point before making the final update allowing it to anticipate the optimal update direction, reducing overshooting and improving convergence.
5. Adam combines advantages of AdaGrad and RMSProp and uses momentum term to accelerate convergence (like Nesterov)

So all our models utilize Adam (Adaptive Movement Estimation) that adjusts the learning rate for each parameter by maintaining exponentially decaying averages of both the first-order (mean) and second-order (uncentered variance) of the gradients. We chose Adam, because it effectively combines the benefits of AdaGrad (which is good for sparse gradients) and RMSPromp (which is good for non-stationary objectives), making it robust for training deep neural networks (Kingma & Ba, 2015). The integration of momentum within Adam, a technique that helps accelerate gradient descent by smoothing out updates applied across training steps,

is significantly beneficial to our models' training efficiency. Instead of reacting solely to the most recent gradient, momentum accumulates a velocity vector pointing in directions that consistently reduce the loss function. This allows the optimizer to navigate ravines and avoid oscillations, leading to faster and more stable convergence(Sutskever et al., 2013).

5.2. Training Dynamics

LEARNING RATE SCHEDULER

ReduceLROnPlateau is introduced after noticing oscillations in training and validation CMaP accuracy in Model 2 (Table 2) after applying SpecAugment, it was intended to help smooth convergence.

- This monitors validation metrics and reduces the learning rate if performance stagnates. It allows for **finer adjustments** to weights, avoiding suboptimal local minima. Instead of updating the learning rate **based on a function of time**, it monitors a **performance metric** (typically validation loss or a custom metric like CMAP@k), and **reduces the learning rate when that metric stops improving**.It is **adaptive in response to training stagnation**, not over epochs. $\eta_{\text{new}} = \eta_{\text{current}} \times$ factor.
- This responds dynamically to validation loss metric (CmAP here), it's reactive and instead of being time dependent on predefined schedule, it is also computationally lighter so we largely opt for this to improve convergence.
- For clarity, the difference from metrics like Adam is that the latter will update every update step and reacts to magnitude of historic gradients, while learning rate schedulers will react to plateauing validation performance.

Cosine Annealing (Loshchilov & Hutter, 2017)

During our training, we adopted a cosine annealing learning rate schedule to better guide convergence during training. Learning rate follows a **cosine decay curve** from an initial maximum to a defined minimum, rather than abruptly dropping it at fixed intervals. This encourages smoother convergence and avoids abrupt jumps in learning rate that can destabilize training.

Small Mathematical Overview

Cosine annealing controls the learning rate as a **deterministic function of time (i.e., epoch or step)**. The learning rate decays from a maximum value to a minimum value using a **cosine function**, giving it a **smooth, non-linear decay profile**:

$$\eta_t = \eta_{\text{min}} + \frac{1}{2}(\eta_{\text{max}} - \eta_{\text{min}}) \left(1 + \cos\left(\frac{T_{\text{cur}}}{T_{\text{max}}}\pi\right)\right)$$

where η_t is the learning step at t, η_{max} and η_{min} are the upper and lower bounds of the LR, T_{cur} is the current step, T_{max} is the total number of steps over which annealing occurs Starts with a high learning rate (encouraging large exploratory steps early). Gradually reduces the learning rate, smoothing convergence as training progresses.

6. Numerical Results

6.1. Dataset Cleaning, Splitting, Processing

We used the BIRDCLEF 2023 dataset from Kaggle, a collection of audio recordings sourced from the xeno-canto community and curated for biodiversity research. Given the diversity and imbalance inherent in ecological datasets, we first applied data cleaning in order to ensure reliable training dropping classes with < 2 datapoints (see Section3). After filtering, 257 bird species remained for model development. To train and evaluate our models we partitioned the cleaned dataset into three splits: Training Set (70%) used for model learning, Validation Set (15%) used to tuned hyperparameters/monitor model performance and Test Set (15%) used for final model evaluation. Stratified Sampling was applied based on species labels to preserve class distribution across splits.

6.2. Feature Extraction and Dataset Organization

Raw audio recordings were transformed into log-mel spectrograms, a standard time-frequency representation for audio classification. Each clip was resampled to a uniform 32 kHz (when necessary). We then trimmed or zero-padded all clips to a fixed duration of five seconds. A mel spectrogram was computed with 128 mel frequency bins, and the resulting power spectrogram was converted to the decibel scale via a logarithmic transformation. This preprocessing pipeline standardizes the input features, enabling stable training across both convolutional or recurrent neural networks. Metadata files were also generated for each dataset split (train, validation, test), containing the path to each spectrogram, its associated bird species label, and encoded class index.

6.3. Numerical Results Goal

The goal of our evaluation is to compare the effectiveness of different architectural decisions and training strategies on the BirdCLEF dataset. We assessed model performance using top-1 accuracy, top-5 accuracy, and CMAP@3, which is a ranking-based metric better suited to imbalanced multiclass settings

6.4. Evaluation Metrics

In our experiments, we tracked both Top-1 and Top-5 accuracy across all models to evaluate performance in a more nuanced manner. Top-1 accuracy measures how often the model's most confident prediction exactly matches the ground truth label. However, given that the BirdCLEF dataset includes over 250 bird species with often acoustically similar calls, Top-1 by itself risks underestimating model competency when near-miss predictions are still plausible. Top-5 accuracy addresses this by considering a prediction correct if the true label is among the model's five most probable guesses. This is important in high-class-count tasks, where some confusion is both expected and tolerable. Therefore, including Top-5 accuracy allows us to obtain a more forgiving and informative metric that better reflects the model's ability to narrow down possibilities in complex environments. This approach is consistent with evaluation strategies in other large-scale classification domains like ImageNet (Petersen et al., 2022). CMAP@3: mean average precision at 3; we used a custom metric that rewards correctly ranked species predictions up to the top 3. This was our primary evaluation metric as it better reflects partial correctness in species classification.This approach aligns well with both applied and theoretical deep learning research, which emphasizes validation loss as a reliable early warning signal for overfitting (CS, 2024).

6.5. Loss Functions

MAE or RMSE is redundant here, bird ID is categorical, not continuous. Categorical cross-entropy works better: it penalizes confident wrong answers and fits the one-hot, mutually exclusive nature of species classification.

$$\text{CCE} = -\sum_{i}^{C} y_i \cdot log(\hat{y}_i)$$

C: number of classes (e.g., bird species)

y_i: ground truth (1 for correct class, 0 otherwise)

$\hat{y}_i$: predicted probability for class i

Since only the correct class contributes (because of one-hot encoding), this simplifies to:

$$\text{CCE} = -\log(\hat{y}_{true})$$

But CCE buckles under class imbalance—common species like Barn Swallow dominate the loss, silencing rarer calls like the Chestnut Sparrow. We expected focal loss to fix this. By downweighting easy examples and applying class weights (α), it could force the model to learn from rare, misclassified species. We set $\gamma = 2.0$ to keep focus tight but training stable however see limited overhead even with oversampling.

6.6. Baseline Methods

Our baseline model used a ResNet50 architecture pretrained on ImageNet, with all convolutional layers frozen. A global average pooling layer, dropout (0.3), and a dense softmax head completed the model, trained with categorical cross-entropy and the Adam optimizer. Despite its simplicity, this setup achieved a validation CMAP@3 of 38%, serving as a solid reference point. However, the model quickly overfitted, with training metrics diverging from validation performance. To address this, we applied SpecAugment (time and frequency masking, Gaussian noise), which significantly reduced overfitting and improved generalization. We also explored EfficientNetB0 as a lightweight alternative, but it did not outperform ResNet50. These early models lacked mechanisms to capture temporal dependencies or handle extreme class imbalance, but they offered valuable baselines for measuring gains from the RNN, attention, and focal loss–based enhancements introduced in later models.

6.7. Summary, Presentation and Discussion of Numerical Results (Table 1

Baseline CMap sits at 34.63% but features massive overfitting across all metrics with $train_CMap$ at 65.62% the model doesn't generalise well to the validation data. Heavy SpecAugment techniques including dynamic time and frequency masking, gaussian noise and time-cropping during preprocessing see overfitting effectively nullified across all future models. Unfreezing just the Conv5D layer doesn't give the model enough time to learn pattern and finetune. So for model 3 we unfreeze Conv3D, 4D, 5D, add an attention head and crucially include a hybrid BiGRU/ BiLTSM to capture temporal variations over long and short ranges (See Table 1). LR on Plateau smooths convergence and this model records a 72% increase over the baseline (and 25% percentage point increase).

A continuation matrix analysis at this stage reveals that the model is ignoring species with lower class sizes which is detrimental to inclusion of rare/ minority species. We hence experiment with focal loss (setting $\gamma = 2$ to account for high class polarisation (175:1) and setting alpha = class weights). However performance lags behind model 3. We further test to see if including focal loss at least improves on minority cases (defined as $<$ Q1 class sizes), but find recall rise for only one bird species Pygmy Batis (ID 165) jumped from 0.000 to 0.500. For the remaining 96%, recall stayed flat at 0.000. The classifier continues to struggle with sparsely represented, acoustically subtle species – likely because there's little pattern diversity for it to lock onto. Finally, we try oversampling rare species in the training set, we do this to increase representation of minority classes to get the model to pay more attention to them. This simple intervention resulted in a significant improvement: 10 rare species, including Buff-bellied Warbler, Chestnut Weaver,

Black Sawwing, Mackinnon's Shrike, and Ross's Turaco, saw recall jump from 0.000 to 0.500 (Figure7) or higher. This is a marked improvement over focal loss alone, which lifted recall for only one class – brown chested Alethe(Table 3). The result suggests that, while focal loss theory prioritizes hard-to-classify examples, it is insufficient on its own when sample counts are severely limited. Oversampling, by increasing exposure, helps the model at least surface pattern diversity for underrepresented species. Still, 45 out of 55 rare species remained undetected- highlighting that data quantity alone doesn't resolve deep overlaps in acoustic structure.

We identify that while there is class imbalance, for completion and rigor of analysis we also look at spectrograms of bird-pairs confused a 100% of the time to ask, does the confusion stem in part from confused species having similar calls? We find extremely high ($> 90\%$) cosine similarity between the species our model is confusing (Table 4). More aggressive $spec_augment$ or training techniques might be and we have exhausted performance upside on ResNET50 we hence pivot to look at different backbones (EfficientNETB03 - 12M parameters) and then pivot into transformer based solutions. Oversampling won't help if two classes are inherently hard to separate in the feature space. Transformers attend across time and frequency in parallel, capturing long-range dependencies from the outset. That means they can pick up on subtle acoustic cues—harmonics, call contours, or trailing echoes—especially valuable when species sound deceptively similar. Unlike CNNs or LSTMs, they don't lose context. They're well-suited for modeling fine structure. While ResNets build their view layer by layer, transformers begin with a global receptive field. They see the whole spectrogram at once – critical when two calls differ not in content, but in structure or timing. Transformer based models also plateau at $\sim$50% this is likely due limitations in class sizes of some species, transformers need large data volumes to perform.

Finally, returning again to the data structure and drilling into possible fixes for data limitations (section 3). The benefits of this approach are two-pronged, we simultaneously access new data on minority classes (note this is different from oversampling that increases representation in training data,, the data multi labelling acquires is fresh spectrogram clips), at the same time we combat rich-feature loss generated when one umbrella 'primary label' is assigned to an audio clip with overlapping bird calls. Though the metadata for secondary calls in audio files is sparse it still sees performance jump to 81%.

6.8. Limitations

Despite strong results, several limitations remain. First, early models used single-label classification by assigning each 5-second spectrogram the primary label from its source file. This overlooked the polyphonic nature of bird recordings, leading to misclassification of overlapping species. Although later models explored multi-label training, this simplification likely suppressed minority signal detection. Second, fixed-length 5s crops – while convenient for training – risk truncating long calls or capturing irrelevant audio. Though SpecAugment and time cropping mitigate this, the rigid segmentation may limit the learning of longer acoustic structures. Third, while SpecAugment reduces overfitting, aggressive masking may remove species-defining frequency bands, particularly in rare or similar-sounding classes. Additionally, class imbalance was only partially addressed via focal loss and class weights; rare classes with < 10 samples remained frequently confused with dominant species. Location metadata (latitude and longitude), though integrated in some models, was not systematically used. This represents a missed opportunity to regularize predictions based on species' geographic distributions. Finally, Transformer-based models showed limited gains, likely due to the dataset's moderate size and training constraints (30 epochs). Transformers generally benefit from large-scale data and longer training schedules, which our setup could not accommodate. Incorporating oversampling could improve performance.

6.9. Future Research

Several promising avenues could enhance both the performance and practical utility of our bird sound classification pipeline. One key direction involves semi-supervised learning methods such as pseudo-labeling or consistency regularization, which can leverage unlabeled or weakly labeled data to improve the representation of rare species without requiring manual annotation. More systematic use of geographic metadata could also play a critical role. By embedding location features or constraining predictions based on regional species distributions, models can better disambiguate acoustically similar birds from different habitats. Temporal modeling could be improved by moving beyond fixed-length crops to overlapping or dynamic cropping strategies, or even full-sequence modeling, thereby preserving the structure of longer vocalizations. Data sampling strategies also warrant refinement; curriculum learning or focal sampling could direct the model's focus toward rare or confusing classes at appropriate stages of training. Further gains may come from optimizing Transformer architectures by exploring audio-specific variants such as the Audio Spectrogram Transformer, Patchout ViT, or CoAtNet, especially when paired with longer training schedules and warmup-based learning rates. Finally, few-shot learning approaches like prototypical or contrastive networks may offer an edge in generalizing to species with very few labeled examples. Together, these directions outline a roadmap toward more robust, context-aware, and field-ready bird call recognition systems.

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Contributions

We predicate this section by reinforcing that multiple parties contributed to a single-section, if not by code, by pointing out misgivings in crucial assumptions and iteratively suggesting modelling improvements. Below, you can find the breakdown on what each member did the most work on, however we have all worked on data preprocessing and all the model implementations, helping each other to check and refine the processes.

- Leah Assefa - Data Preprocessing lead
- Sharanya Seth - SpecAugment, ResNET50 models (CNN-RNN and hybrid), confusion matrix, focal loss and oversampling to tackle class imbalance.
- David Vardanyan - EfficientNET models, feature extraction and model refinement.
- Weichen Wang - Transformers and Multimodal models, identification and mining of metadata.

You can find the notebook via the link: Click here

Appendix

Model	Comments	Train_Top-1 Accuracy	Train_CMAP@3	Train_Top-5 Accuracy	Val_Top-1 Accuracy	Val_CMAP@3	Val_Top-5 Accuracy
Model 0 : ResNet50 Frozen (Baseline Model)	The baseline CMap is 34.63%. Though train_CMaP is almost 50% higher indicating a gross overfitting	~58.68%	~65.62 %	~79.95%	~27.78%	~34.63 %	~49.%
Model 1 : ResNet50 + SpecAugment	We record a ~5% decrease in val_CMap. However validation outperforms training so **SpecAug is highly effective in mitigating overfitting.** LR on plateau introduced for better convergence	~15.41%	~20.70 %	~35.32% %	~21.12%	~27.84 %	~42.56%
Model 2 : ResNet50 + conv5 Fine-Tuned	We see a drop in validation metrics across the board, we likely unfroze too few layers. Positively, however val_df stil outperforms train_df so specaugment effective in combating overfitting.	~11.76%	~16.44 %	~29%	~16.58%	~22.27 %	~36.32%
Model 3 : ResNet50 + BiGru-BiLSTM + Attention	**Adding sequential architecture and a self attention head sees 72% improvement from baseline (34.6% to 59.56%). Overfitting controlled.**	~60.45%	~66%	~77.39%	~53.38%	~59.56 %	~72.44%
Model 4 : ResNet50 conv3/4/5 + Focal Loss	Confirmation matrix reveals class imbalance, so we introduce focal loss. However val metrics drop. Model also performs better at categorising only 1 minority species. Overfitting remains well controlled.	48.10%	53.58%	65.61%	46.35.%	52.75%	65.97%
Model 4.5 : ResNet50 conv3/4/5 + Focal Loss + **Oversampling**	Lastly, to deter class imbalance we oversample rare species from training data. This sees CMap rise to 63.18% a marked improvement on pure focal loss and a rise from Model3 though architecture is held constant. Meanwhile cosine similarity between spectrograms of confused bird pairs is extremely high so we pivot to transformer architectures.	73.98%	77.74%	85.36%	57.56%	63.18%	75.52%
Model 5: EfficientNetB0 Baseline	Perhaps ResNET50 had too few parameters to finetune on, so we experimented with EfficientNETB03 but baseline performance isn't promising.	6.36%	9.86%	21.11%	8.65%	12.62%	24.20%
Model 6 : EfficientNetB3 + BiLSTM + Attention + Focal	Building on what we learnt to unfreeze higher layers, add self-attention heads and add hybrid BiGRU/ BiLTSM.	19.32%	24.64%	37.74%	27.40%	32.91%	44.45%
Model 7 : ViT - Transformer	Transformers need huge amount of data and our polarised df with class size <3 may impede performance. H-Acc also reported climbs to 99.54%	-%	50.10%	-%	-%	50.07% %	-%
Model 8 : Multilabelling	Building on what we learnt - we mine metadata to shift to multi label classification. This reduced class polarisation by introducing new data for rare species while also fixing the single label for rich audio file fallacy. Allowing for richer feature mapping sees CMap performance jump to ~81%	-	80.84%	-	-	81.27%	-%

Table 1. Summary of Findings

True Species	Predicted As	Confusion Rate
African Gray Flycatcher	Cinnamon Bracken-Warbler	100.0%
Black-and-white Mannikin	Barn Swallow	100.0%
Black-throated Apalis	African Thrush	100.0%
Black-throated Barbet	Black Crake	100.0%
Kikuyu White-eye	Gray-backed Camaroptera	100.0%
Buff-bellied Warbler	Red-backed Scrub-Robin	100.0%
Chestnut Sparrow	Common House-Martin	100.0%
Chestnut Weaver	Emerald-spotted Wood-Dove	100.0%
African Darter	Sulphur-breasted Bushshrike	100.0%
Dusky Turtle-Dove	Mourning Collared-Dove	100.0%
Equatorial Akalat	Chinspot Batis	100.0%
Gray-throated Barbet	Gray-backed Camaroptera	100.0%
Hinde's Pied-Babbler	Woodland Kingfisher	100.0%
Joyful Greenbul	African Goshawk	100.0%
Eastern Violet-backed Sunbird	Red-faced Crombec	100.0%

Species Sample Counts:

Eastern Violet-backed Sunbird	6	samples
African Gray Flycatcher	6	samples
Mourning Collared-Dove	26	samples
African Goshawk	40	samples
African Thrush	31	samples
Barn Swallow	350	samples
Black-and-white Mannikin	5	samples
Black Crake	42	samples
Black-throated Apalis	8	samples
Black-throated Barbet	5	samples
Red-backed Scrub-Robin	197	samples
Red-faced Crombec	29	samples
Kikuyu White-eye	6	samples
Buff-bellied Warbler	6	samples
Chestnut Sparrow	4	samples
Chestnut Weaver	6	samples
Chinspot Batis	55	samples
Cinnamon Bracken-Warbler	79	samples
Common House-Martin	297	samples
African Darter	5	samples
Dusky Turtle-Dove	6	samples
Equatorial Akalat	5	samples
Emerald-spotted Wood-Dove	44	samples
Sulphur-breasted Bushshrike	63	samples
Gray-backed Camaroptera	167	samples
Gray-throated Barbet	6	samples
Woodland Kingfisher	47	samples
Hinde's Pied-Babbler	6	samples
Joyful Greenbul	5	samples

Table 2. Species with 100% Confusion Rate

Class ID	Recall M3	Recall M4	Delta	Species	
7	0.000	0.000	0.000	African Gray Flycatcher	6
18	0.000	0.000	0.000	Augur Buzzard	8
22	0.000	0.000	0.000	Black-and-white Mannikin	5
36	0.000	0.000	0.000	Black Sawwing	9
39	0.000	0.000	0.000	Black-throated Apalis	8
40	0.000	0.000	0.000	Black-throated Barbet	5
43	0.500	0.500	0.000	Brown-chested Alethe	7

Table 3. Per-Class Recall for Rare Species

True Class	Predicted As	Cosine Similarity
7	69	0.9670
22	98	0.9700
39	16	0.8749
40	90	0.9774
51	97	0.9145
57	162	0.8591
64	20	0.9411
65	88	0.8185
79	105	0.9762
82	9	0.9812
87	174	0.8880
117	97	0.9462
122	56	0.9674
126	10	0.9107
129	64	0.8467

True Species	Predicted As	Cosine Similarity
African Gray Flycatcher	Cinnamon Bracken-Warbler	0.9670
Black-and-white Mannikin	Green-headed Sunbird	0.9700
Black-throated Apalis	African Thrush	0.8749
Black-throated Barbet	Fan-tailed Raven	0.9774
Kikuyu White-eye	Gray-backed Camaroptera	0.9145
Buff-bellied Warbler	Pied Kingfisher	0.8591
Chestnut Sparrow	Barn Swallow	0.9411
Chestnut Weaver	Emerald-spotted Wood-Dove	0.8185
African Darter	Great Cormorant	0.9762
Dusky Turtle-Dove	Mourning Collared-Dove	0.9812
Equatorial Akalat	Red-chested Cuckoo	0.8880
Gray-throated Barbet	Gray-backed Camaroptera	0.9462
Hinde's Pied-Babbler	Brown-throated Wattle-eye	0.9674
Joyful Greenbul	African Paradise-Flycatcher	0.9107
Eastern Violet-backed Sunbird	Chestnut Sparrow	0.8467

Table 4. Cosine Similarity Table

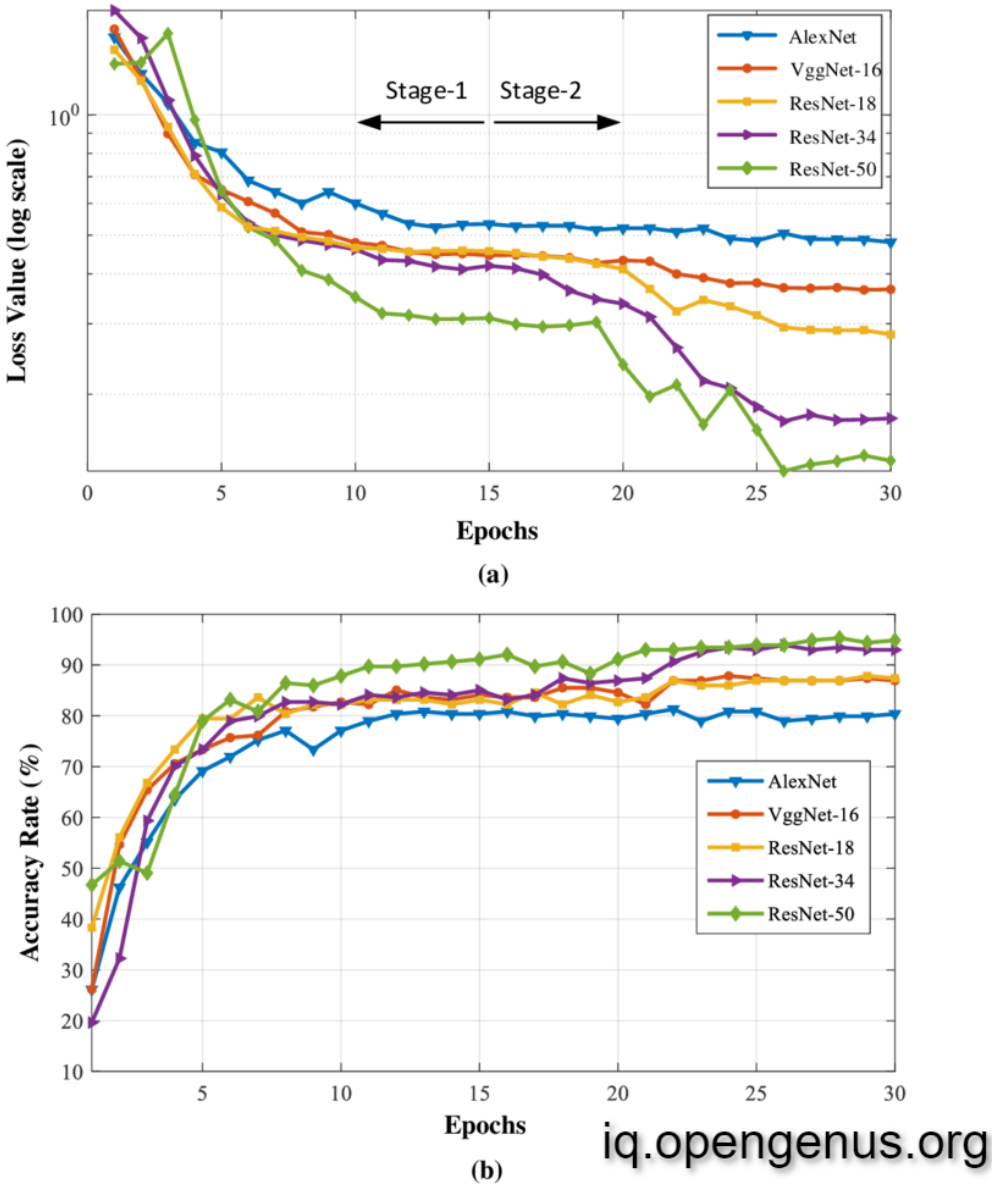


Figure 1. Training Curves: Loss and Accuracy Across Epochs for AlexNet, VGG, and ResNet Models

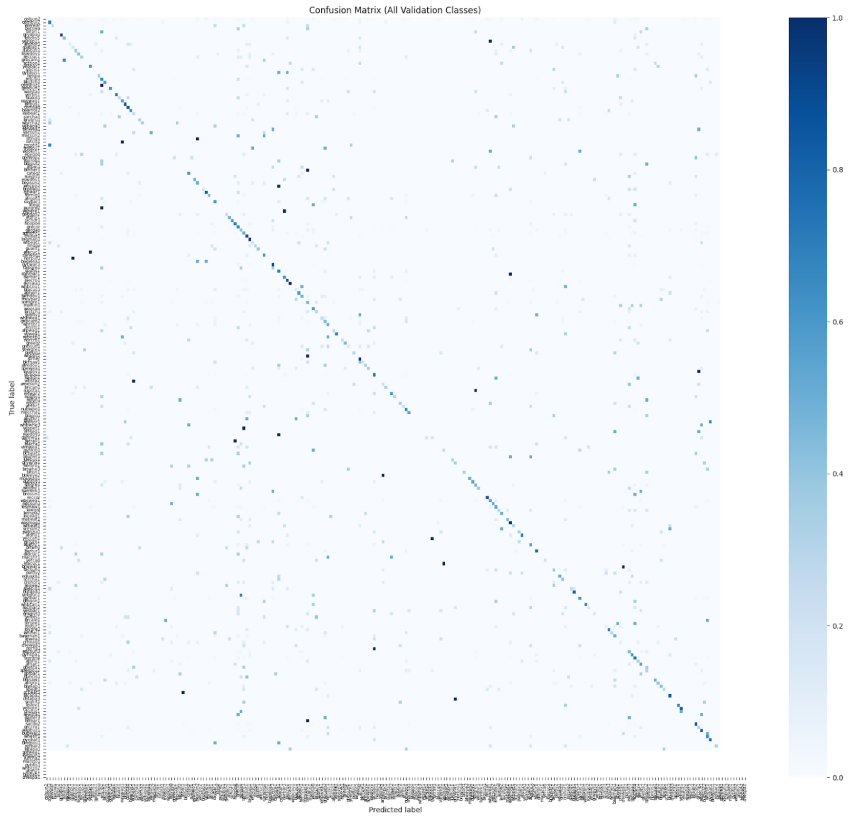


Figure 2. Confusion Matrix: All Validation Classes

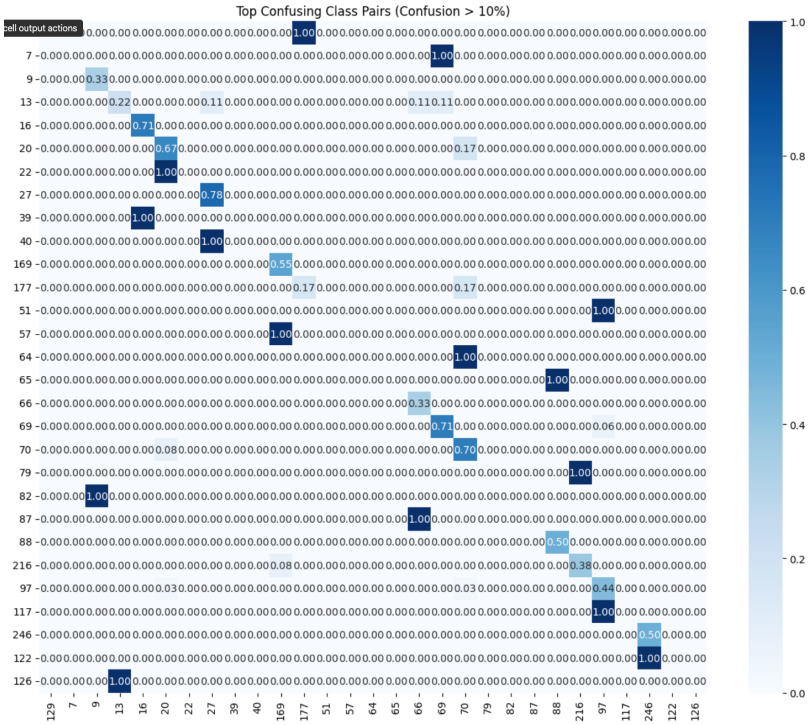


Figure 3. Top Confusing Class Pairs (>10%)

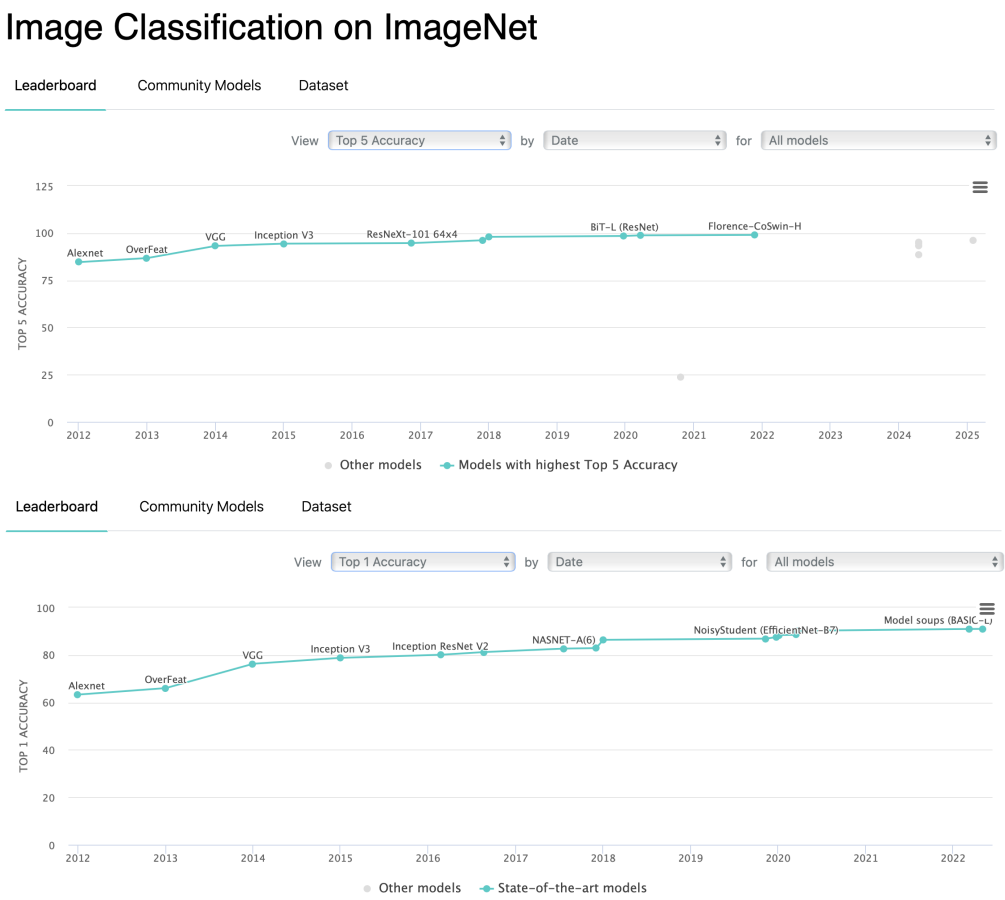


Figure 4. Image Net Top-1/Top-5 Accuracy Over Time

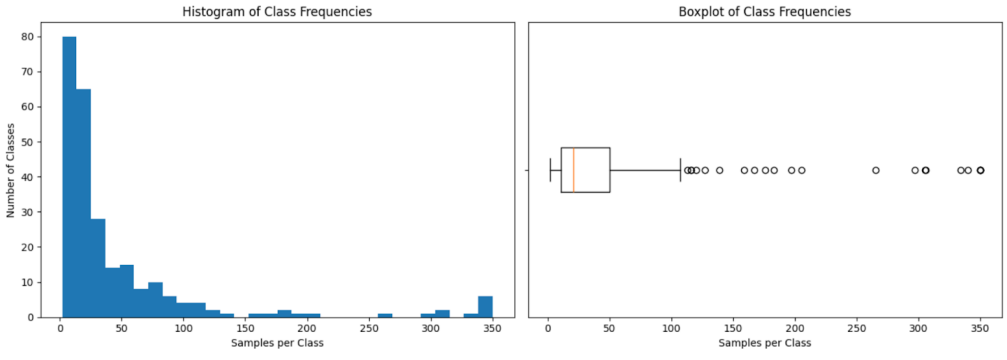


Figure 5. Histogram + Boxplot of Class Frequencies

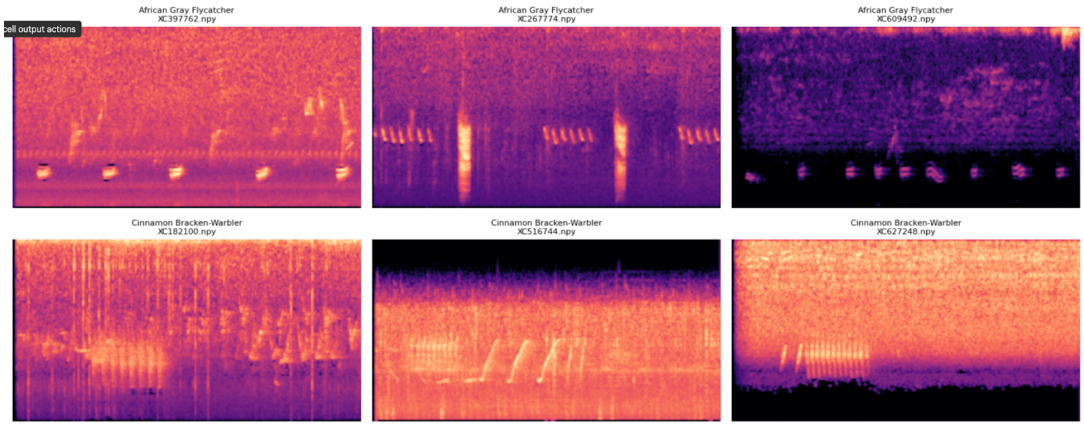


Figure 6. Spectrogram Comparisons Between Confused Pairs

Class ID	Recall M3	Recall M4.5	Delta	Species	Count
7	0.000	0.000	0.000	African Gray Flycatcher	6
18	0.000	0.000	0.000	Augur Buzzard	8
22	0.000	0.000	0.000	Black-and-white Mannikin	5
36	0.000	1.000	1.000	Black Sawwing	9
39	0.000	1.000	1.000	Black-throated Apalis	8
40	0.000	0.000	0.000	Black-throated Barbet	5
43	0.000	0.500	0.500	Brown-chested Alethe	7
44	0.000	0.000	0.000	Bristle-crowned Starling	2
51	0.000	0.000	0.000	Kikuyu White-eye	6
57	0.000	1.000	1.000	Buff-bellied Warbler	6
63	0.000	0.500	0.500	Cinnamon-chested Bee-eater	9
64	0.000	0.000	0.000	Chestnut Sparrow	4
65	0.000	1.000	1.000	Chestnut Weaver	6
79	0.000	0.000	0.000	African Darter	5
81	0.000	0.000	0.000	Double-toothed Barbet	2
82	0.000	0.000	0.000	Dusky Turtle-Dove	6
87	0.000	0.000	0.000	Equatorial Akalat	5
91	0.000	0.000	0.000	Fan-tailed Widowbird	4
100	0.000	0.000	0.000	Golden-breasted Starling	2
101	0.000	0.000	0.000	Golden-backed Weaver	4
104	0.000	0.000	0.000	Gray Crowned-Crane	8
111	0.000	0.000	0.000	Gray-backed Fiscal	7
117	0.000	0.000	0.000	Gray-throated Barbet	6
122	0.000	0.000	0.000	Hinde's Pied-Babbler	6
125	0.000	0.000	0.000	Hunter's Sunbird	4
126	0.000	0.000	0.000	Joyful Greenbul	5
129	0.000	0.000	0.000	Eastern Violet-backed Sunbird	6
132	0.000	0.000	0.000	Lesser Masked-Weaver	10
140	0.000	0.000	0.000	Long-toed Lapwing	2
143	0.000	1.000	1.000	Mackinnon's Shrike	4
145	0.000	0.000	0.000	Marabou Stork	4
147	0.000	0.000	0.000	Mouse-colored Penduline-Tit	6
158	0.000	0.000	0.000	Parrot-billed Sparrow	5
159	0.000	0.500	0.500	Pale Flycatcher	10
160	0.000	0.000	0.000	Pale Prinia	3
164	0.000	0.000	0.000	Purple Grenadier	7
165	0.000	0.000	0.000	Pygmy Batis	10
176	0.000	0.500	0.500	Red-fronted Barbet	8
179	0.000	0.000	0.000	Red-fronted Prinia	6
181	0.000	0.000	0.000	Red-headed Weaver	4
187	0.000	0.500	0.500	Ross's Turaco	7
189	0.000	0.000	0.000	Rufous Chatterer	4
190	0.000	0.000	0.000	African Sacred Ibis	4
194	0.000	0.000	0.000	Shelley's Starling	2
208	0.000	0.000	0.000	Spot-flanked Barbet	9
209	0.000	0.000	0.000	Speckle-fronted Weaver	8
215	0.000	0.000	0.000	Stuhlmann's Starling	3
218	0.000	0.000	0.000	Tacazze Sunbird	4
232	0.000	0.000	0.000	White-bellied Canary	7
234	0.000	0.000	0.000	White-browed Crombec	5
238	0.000	0.000	0.000	White-chinned Prinia	6
242	0.000	0.000	0.000	White Helmetshrike	10
243	0.000	0.000	0.000	White-rumped Shrike	6

Figure 7. Rare Species Recall Comparison Between Models 3 and 4.5 (With Oversampling)